

Chapter 6

Pricing Long Dated Products via Libor Approximations

6.1 Introduction

In this chapter we study lognormal approximations for Libor market models, proposed in Kurbanmuradov, Sabelfeld and Schoenmakers (2002), where special attention is paid to their simulation by direct methods and lognormal random fields. Generally, the main advantage of log-normal approximations is that their distributions can be simulated fast since, in contrast to the usual numerical solution of the Libor SDE, the approximations can be simulated directly at any future point in time. For instance, the lognormal approximation proposed in Brace, Gatarek and Musiela (1997) can be simulated effectively by a Gaussian random field of log-Libors. As a result, since in general valuation of Libor derivatives comes down to computation of expected values of functions of Libors, an important family of Libor instruments consisting of long dated exotic products can be valued faster by using lognormal approximations. As such the proposed approximations provide valuable alternatives to the Euler method, in particular for long dated instruments.

After introducing different lognormal Libor approximations we carry out a path-wise comparison with the “exact” SDE solution obtained by the Euler scheme using sufficiently small time steps. Also we test approximations obtained via numerical solution of the SDE by the Euler method, using larger time steps. It turns out that, for typical volatilities observed in practice, improved versions of the Brace, Gatarek and Musiela (1997) approximation appear to have excellent path-wise accuracy. Although this accuracy can be achieved also by Euler stepping the SDE using larger time steps as we will see, it follows from a comparative cost analysis that, particularly for long maturity options, the Euler method is more time consuming than the lognormal approximation. We illustrate this at some example Libor options with 12 yr. maturity.

In Section 6.2 we derive different lognormal approximations for a Libor market model in the terminal bond measure given by SDE (1.16) and study their accuracy. In particular, the lognormal approximations are subjected to a mutual path-wise comparison with approximations obtained by Euler stepping

the SDE. A ranking between the different approximations is thus obtained. Then, in Section 6.3, we construct and investigate direct simulation techniques for the approximations derived in Section 6.2. The efficiency of the lognormal approximations with respect to Euler stepping the SDE is analysed in Section 6.4. Based on this analysis we propose an “optimal” simulation program in which different approximations may be combined. In Section 6.5 we consider the valuation of swaptions and trigger swaps for different approximations and compare the results. Also we discuss briefly the callable reverse floater, an exotic instrument for which the proposed simulation method is extremely efficient.

Throughout this chapter we work with the Libor market model in the terminal measure (1.16). It will be clear, however, how to convert the presented Libor approximations to a Libor SDE in another measure, and that generally the same conclusions as in Section 6.6 can be made. For the experiments in this chapter we used a model with time independent scalar volatilities and time independent instantaneous correlations due to a correlation generating sequence of the form (2.48),

$$b_i = \exp[\beta(i-1)^\alpha], \quad \beta > 0, \quad 0 < \alpha < 1, \quad (6.1)$$

$$\rho_{ij}(t) \equiv \frac{\min(b_i, b_j)}{\max(b_i, b_j)} = \exp[-\beta|(i-1)^\alpha - (j-1)^\alpha|], \quad 1 \leq i, j < n.$$

This structure is more simple than (2.46) for instance, but it has full rank and is of ratio form (2.29) which satisfies (2.30). So, the structure (6.1) has realistic features yet. In particular, the condition $0 < \alpha < 1$ implies that instantaneous correlations $\rho(dL_i, dL_{i+p})$ are increasing in i for fixed p , which is very natural indeed.

6.2 Different Lognormal Approximations

Let us recall the Libor SDE (1.16) in the terminal measure P_n ,

$$dL_i = - \sum_{j=i+1}^{n-1} \frac{\delta_j L_i L_j}{1 + \delta_j L_j} \gamma_i^\top \gamma_j dt + L_i \gamma_i^\top dW^{(n)}, \quad t_0 \leq t \leq T_i, \quad (6.2)$$

where $0 \leq t_0 \leq T_i$ and $1 \leq i < n$. It is convenient to deal with (6.2) in the following integrated form,

$$\ln \frac{L_i(t)}{L_i(t_0)} = - \int_{t_0}^t \sum_{j=i+1}^{n-1} \frac{\delta_j L_j}{1 + \delta_j L_j} |\gamma_i| |\gamma_j| \rho_{ij} ds - \frac{1}{2} \int_{t_0}^t |\gamma_i|^2 ds + \int_{t_0}^t \gamma_i^\top dW^{(n)}, \quad (6.3)$$

where $\rho_{ij} = \gamma_i^\top \gamma_j / |\gamma_i| |\gamma_j|$. In practice, we may define the vectors $\gamma_i / |\gamma_i|$ through the matrix (ρ_{ij}) by applying a Cholesky decomposition.

Note that only the first term in the right hand side of (6.3) is generally non-Gaussian. Let us consider the contribution of the non-Gaussian term where we assume for simplicity that the functions γ_i are constants. We introduce the notations: $\rho_i = \sum_{j=i+1}^{n-1} |\rho_{ij}|$, $\rho = \max_i \rho_i$, $\delta = \max_i \delta_i$, and $\gamma = \max_i |\gamma_i|$.

Let us denote by $\tilde{L}$ the maximum value of the L_i , i.e., $\tilde{L} = \max_i \sup_{t_0 \leq t \leq T_i} L_i(t)$.

Then, we may write (6.3) as

$$\ln \frac{L_i(t)}{L_i(t_0)} = \varepsilon_i - \frac{1}{2} |\gamma_i|^2 (t - t_0) + |\gamma_i| \sqrt{t - t_0} Z_i(t),$$

where $Z_i(t)$ is a standard normally distributed random variable and ε_i can be estimated by $|\varepsilon_i| \leq (t - t_0) \delta \tilde{L} \gamma^2 \rho_i$. So, by neglecting ε_i we cause in L_i only a small *relative* error of order of ε_i when

$$|\varepsilon_i| \leq (t - t_0) \delta \tilde{L} \gamma^2 \rho_i \leq (t - t_0) \delta \tilde{L} \gamma^2 \rho \ll 1. \quad (6.4)$$

Note that for typical values, e.g., $\delta = 0.25$, $\gamma = 0.15$, $\tilde{L} = 0.07$, $t - t_0 = 10$, this relative error is about 0.4%. However, dependent on ρ and the length of the tenor structure this error can become rather large in practice.

The approximation by neglecting the non-Gaussian terms ε_i in (6.3) will be called (0)-approximation to (6.2). In this approximation, forward Libor rates satisfy

$$dL_i^{(0)} = L_i^{(0)} \gamma_i^\top dW^{(n)}, \quad (6.5)$$

and are given by the explicit solution

$$L_i^{(0)}(t) = L_i(t_0) \exp \left\{ -\frac{1}{2} \int_{t_0}^t |\gamma_i|^2(s) ds + \int_{t_0}^t \gamma_i(s)^\top dW^{(n)}(s) \right\}. \quad (6.6)$$

In Figure 6.1 and Figure 6.2 we show for illustration some samples of $L_i(t)$ and $L_i^{(0)}(t)$. For these figures, scalar volatilities are taken to be relatively high, $|\gamma_1| = \dots = |\gamma_{n-1}| = 0.4$, with $n = 31$, in order to amplify effects. The correlation matrix ρ_{ij} is given by the two parametric structure (6.1), where $\alpha = 0.8$, $\beta = 0.1$ in Figure 6.1, and $\alpha = 0.8$, $\beta = 0.3$ in Figure 6.2. Further, $t_0 = 0$ and the tenor structure is given by $T_i = i\delta$ with $\delta = 0.25$, $i = 1, \dots, 31$. The initial L values are taken to be $L_i(0) = 0.061$. The “true” process $L_i(t)$ is simulated by an Euler scheme with time discretization step $\delta/10$.

From the trajectories in Figure 6.1 and Figure 6.2 we see that on an initial time interval, the trajectory $L_{10}^{(0)}$ approximates the trajectory L_{10} very well. For increasing time, however, the discrepancy increases. Also, in accordance with (6.4), for increasing ρ in (6.4), respectively decreasing β in (6.1), this

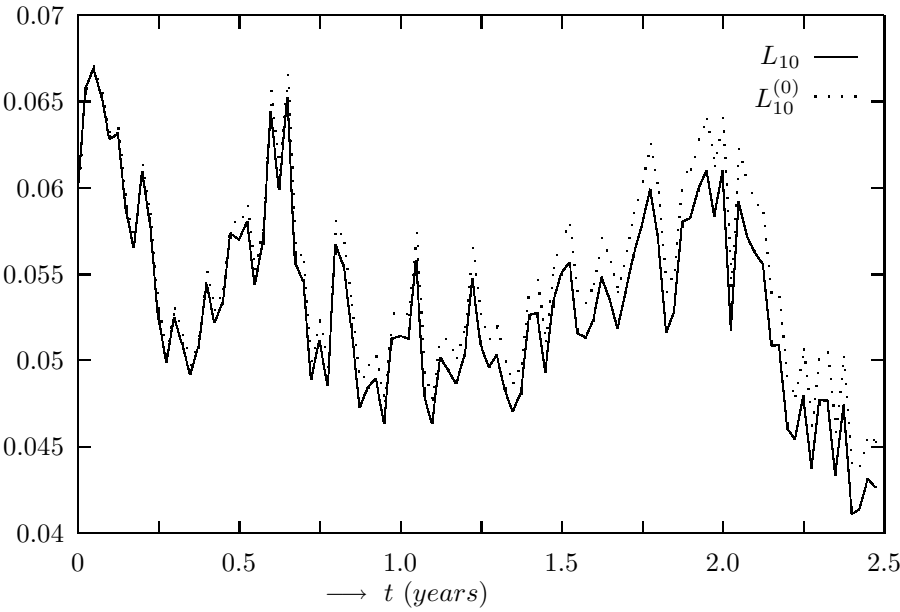


FIGURE 6.1: Sample of $L_{10}(t)$ and $L_{10}^{(0)}(t)$, $\alpha = 0.8$ and $\beta = 0.1$

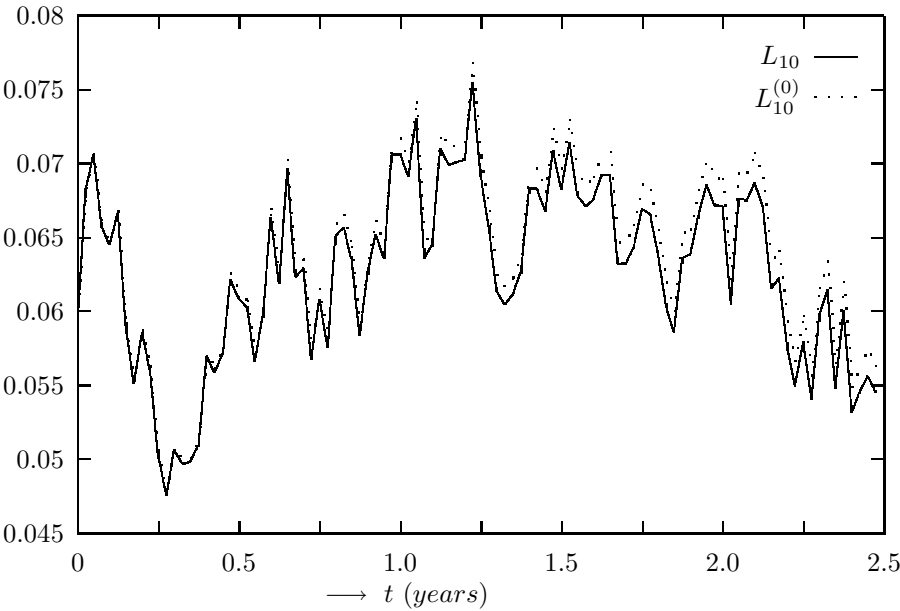


FIGURE 6.2: Sample of $L_{10}(t)$ and $L_{10}^{(0)}(t)$, $\alpha = 0.8$ and $\beta = 0.3$

discrepancy increases. These effects are illustrated by Figure 6.1 and Figure 6.2 and their comparison. So, on the one hand, we see from the plots in Figures 6.1 and 6.2 that, in accordance with (6.4), the (0)-approximation performs well for small times. On the other hand, we see from (6.4) that the (0)-approximation produces good results also for large i , since ρ_i decreases with i (in particular, ρ_{n-1} vanishes). More details about the (0) and other approximations are presented in Tables 6.2–6.5.

In Figure 6.3 we show a sample of the bond price $B_{31}(T_i)$ and its (0)-approximation $B_{31}^{(0)}(T_i)$, $i = 0, \dots, 31$, for the same model parameters as used for Figure 6.1. The bond prices are computed from the Libor process, via the relationship

$$B_{31}(T_i) = \prod_{j=i}^{30} (1 + \delta L_j(T_i))^{-1}, \quad (6.7)$$

and a similar one for the (0)-approximations, where for the “true” bond prices we have used “true” Libors, simulated by an Euler scheme with time discretization step $\delta/10$.

In contrast to the results presented in Figures 6.1 and 6.2, the maximum discrepancy in Figure 6.3 occurs around the middle of the time interval $(0, T_{31})$. The reason is clear from (6.7). Indeed, when i is either close to zero or when it is close to 30, where the drift terms become small, the approximations $L_j^{(0)}(T_i)$, $j = i, \dots, 30$ are close to $L_j(T_i)$ and so $B_{31}^{(0)}(T_i)$ is close to $B_{31}(T_i)$. Note that, exactly, $B_{31}^{(0)}(30 \times 0.25) = B_{31}(30 \times 0.25)$ because L_{30} and $L_{30}^{(0)}$ coincide in the terminal measure given by $n = 31$.

6.2.1 More Lognormal Approximations

It is of interest to consider more refined approximations to L and in particular to look for lognormal approximations improving $L^{(0)}$. So, in fact, we need to find a deterministic or normal approximation to the sum term in the integral representation (6.3). Therefore, for each j we approximate the process

$$Z_j(t) := \frac{\delta_j L_j}{1 + \delta_j L_j} \quad (6.8)$$

in the following way. Let the function f be defined as $f(x) := x/(1+x)$ and $Z_j = f(\delta_j L_j)$. Hence the process $Z := [Z_1, \dots, Z_{n-1}]$ satisfies the SDE system

$$\begin{aligned} dZ_j &= f'(\delta_j L_j) \delta_j L_j \gamma_j^\top dW^{(n)} + \frac{1}{2} f''(\delta_j L_j) (\delta_j L_j |\gamma_j|)^2 dt \\ &\quad - \sum_{k=j+1}^{n-1} \frac{\delta_j \delta_k L_j L_k \gamma_j^\top \gamma_k}{1 + \delta_k L_k} f'(\delta_j L_j) dt \\ &= : a_j(Z, t) dt + b_j(Z, t)^\top dW^{(n)}(t), \quad 1 \leq j < n, \end{aligned} \quad (6.9)$$

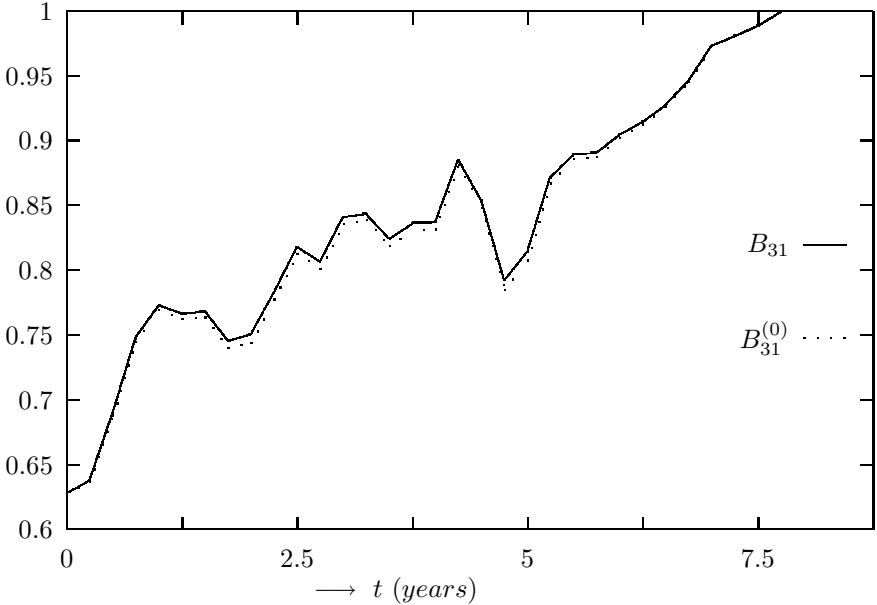


FIGURE 6.3: Sample of $B_{31}(t)$ and its (0)-approximation $B_{10}^{(0)}(t)$, same parameters as for Figure 6.1

with initial conditions $Z_j(t_0) = f(\delta_j L_j(t_0))$. Note that by solving (6.8) for the L_j 's and substituting the result in (6.9) it is immediately clear how the coefficients a_j and b_j should be defined explicitly.

The Picard-Lindelof–0 and Picard-Lindelof–1 iteration for the solution of this SDE are

$$\begin{aligned}
 Z_j^{(0)}(t) &\equiv Z_j(t_0) = \frac{\delta_j L_j(t_0)}{1 + \delta_j L_j(t_0)} \quad \text{and} \\
 Z_j^{(1)}(t) &:= Z_j(t_0) + \int_{t_0}^t [a_j(Z^{(0)}(t_0), s)ds + b_j(Z^{(0)}(t_0), s)^\top dW^{(n)}(s)] \\
 &= f(\delta_j L_j(t_0)) + \frac{1}{2} f''(\delta_j L_j(t_0)) \delta_j^2 L_j^2(t_0) \int_{t_0}^t |\gamma_j|^2 ds \\
 &\quad - \sum_{k=j+1}^{n-1} \frac{\delta_j \delta_k L_j(t_0) L_k(t_0)}{1 + \delta_k L_k(t_0)} f'(\delta_j L_j(t_0)) \int_{t_0}^t \gamma_j^\top \gamma_k ds \\
 &\quad + f'(\delta_j L_j(t_0)) \delta_j L_j(t_0) \int_{t_0}^t \gamma_j^\top dW^{(n)}, \quad 1 \leq j < n, \quad (6.10)
 \end{aligned}$$

and so are deterministic and Gaussian, respectively. The next Picard-Lindelof iteration, however, will be non-Gaussian in general.

By using the approximations $Z_j^{(0)}$ for the expression (6.8) in (6.3), we find

a lognormal approximation which we call the (g) -approximation,

$$\begin{aligned} \ln \frac{L_i^{(g)}(t)}{L_i(t_0)} &:= - \int_{t_0}^t \frac{|\gamma_i(s)|^2}{2} ds + \int_{t_0}^t \gamma_i^\top(s) dW^n(s) \\ &\quad - \sum_{j=i+1}^{n-1} \int_{t_0}^t \frac{\delta_j L_j(t_0) \gamma_i^\top \gamma_j(s)}{1 + \delta_j L_j(t_0)} ds \end{aligned} \quad (6.11)$$

which turns out to be a considerable path-wise improvement of the (0) -approximation and is suggested in Brace, Gatarek & Musiela (1997). See also Schoenmakers & Coffey (1998) for several applications, for instance, a multi-factor generalization of the swaption approximation in Brace, Gatarek & Musiela (1997).

By expanding f, f' , and f'' as $f(x) = x + \mathcal{O}(x^2)$, $f'(x) = 1 + \mathcal{O}(x)$, and $f''(x) = -2 + \mathcal{O}(x)$, respectively and denoting identity modulo terms of order $\mathcal{O}(\delta_j^2 L_j^2(t_0))$ by $\cong$, we have

$$\begin{aligned} Z_j^{(1)}(t) &\cong \delta_j L_j(t_0) \left(1 + \int_{t_0}^t \gamma_j^\top dW^{(n)} \right) \\ &\quad - \sum_{k=j+1}^{n-1} \frac{\delta_j \delta_k L_k(t_0) L_j(t_0)}{1 + \delta_k L_k(t_0)} \int_{t_0}^t \gamma_j^\top \gamma_k ds =: \tilde{Z}_j(t). \end{aligned} \quad (6.12)$$

We note that, while in (6.12) the terms in the sum are of order $\mathcal{O}(\delta_j^2 L_j^2(t_0))$, their sum is possibly of order $\mathcal{O}(\delta_j L_j(t_0))$ and therefore not neglected. Now, by substituting $\tilde{Z}$ for (6.8) in (6.3) we get another lognormal approximation, the $(g1)$ -approximation,

$$\ln \frac{L_i^{(g1)}(t)}{L_i(t_0)} = - \sum_{j=i+1}^{n-1} \int_{t_0}^t \tilde{Z}_j(s) \gamma_i^\top \gamma_j(s) ds - \frac{1}{2} \int_{t_0}^t |\gamma_i(s)|^2 ds + \int_{t_0}^t \gamma_i^\top(s) dW^{(n)}(s). \quad (6.13)$$

The $(g1)$ -approximation in its turn improves the (g) -approximation significantly as will appear from a comparative analysis in (6.2.2). We will also consider a simplification of the $(g1)$ -approximation, the $(g1')$ -approximation, which is obtained by dropping the sum-term in (6.12).

Finally, by substituting $Z^{(1)}$ for (6.8) in (6.3), so without linearization of the function f and its first and second derivative, we get a further refinement, the $(g2)$ -approximation, given by

$$\ln \frac{L_i^{(g2)}(t)}{L_i(t_0)} = - \sum_{j=i+1}^{n-1} \int_{t_0}^t Z_j^{(1)}(s) \gamma_i^\top \gamma_j(s) ds - \frac{1}{2} \int_{t_0}^t \gamma_i^2(s) ds + \int_{t_0}^t \gamma_i^\top(s) dW^{(n)}(s), \quad (6.14)$$

where $Z^{(1)}$ is given by (6.10).

6.2.2 Simulation Analysis of Different Libor Approximations

It will now be interesting to carry out a comparative numerical analysis of the different lognormal approximations. For typical market volatilities we will compare the lognormal approximations path-wise with the “true” solution, obtained by solving the stochastic differential equation (6.2) by the Euler method using small time steps. In addition, we will carry out a comparison of (non-lognormal) approximations obtained by the Euler scheme using larger time steps. For a uniform tenor structure we will experiment with Euler steps equal to δ , 2δ , or 3δ , etc.

For a correct path-wise comparison, we construct all the lognormal approximations in one common probability space by solving their accompanying stochastic differential equations by the Euler scheme with small time steps, namely, the time steps used for the “true” solution. In the numerical schemes, this is easily achieved by using one and the same Wiener increments for all approximations. In fact, we will solve the SDE’s for the log-Libors and their approximations rather than the Libors as in this way we attain a path-wise accuracy of order *one* with the Euler scheme, due to the deterministic diffusion coefficients in the several log – Libor SDE’s. For an extensive study of the numerical solution of stochastic differential equations we refer to the works of Kloeden & Platen (1992), Milstein (1995), and Milstein & Tret'yakov (2004). The different SDE’s for the log – Libors are straightforwardly obtained by taking the differential form of (6.11), (6.13), and (6.14), respectively.

In our next experiments we take $n = 61$ and $\delta_i = 0.25$, i.e., a rather large tenor structure of fifteen years, along with the same initial conditions as used for the generation of Figures 6.1 and 6.2: $L_i(0) = 0.061$. We take for European markets realistic scalar volatilities, $|\gamma_i| = 0.15$, and for the correlation structure (6.1) we take $\alpha = 0.9$ and $\beta = 0.04$, yielding a more or less realistic correlation matrix (Table 6.1). For comparison see, for instance, the correlation table in Brace, Gatarek & Musiela (1997). Note that the off-diagonals in Table 6.1 are slightly increasing. The “true” solution is simulated by the Euler method using $\Delta t = \delta/5$.¹ In Tables 6.2– 6.5 it is shown how often the relative error (in per cent) of the corresponding approximation to L_{12} , L_{24} , L_{36} , and L_{48} lays in the relevant percentage intervals (there is no table for L_{60} because, of course, L_{60} is exactly lognormal in the terminal measure). The relative error between $\hat{L}_i$, the numerical solution to the original equation (6.2) and, for instance $\hat{L}_i^{(g)}$, the numerical solution to the SDE belonging to (g)-approximation is defined as

$$\epsilon_i^{(g)} = \max_{1 \leq j \leq i} \frac{|\hat{L}_i(T_j) - \hat{L}_i^{(g)}(T_j)|}{\hat{L}_i(T_j)}. \quad (6.15)$$

¹For this choice of Δt it appeared by comparison with much smaller Δt that the discretization bias of the “true” solution is negligible w.r.t. the bias of the different approximations.

The relative errors to other approximations are defined analogously.

The numbers in the columns 2 - 8 of Table 6.2 show the number of samples out of 100 for which the event shown in the first column happens. First, we conclude that the path-wise errors produced by the (0)-approximation are generally not tolerable as almost every path has an error in the sense of (6.15) of at least 2-3%. The (g)-approximation proposed by Brace, Gatarek & Musiela (1997) and used for their swaption approximation formula there gives a considerable improvement but is clearly outperformed by the lognormal approximations ($g1'$), ($g1$), and ($g2$). For smaller to moderate ϵ s, ($g2$) appears to perform slightly better than ($g1$) and, in its turn, ($g1$) somewhat better than ($g1'$), as expected; however, for high ϵ (close to 100%) the reverse conclusion can be made, which indicates that ($g2$) produces higher outliers than ($g1$) and so on. Obviously, the (non-lognormal) δ -stepping Euler approximation appears to be the most accurate, but, as we will see in Section 6.3, in many applications it is more expensive than the proposed lognormal approximations since the distributions of the latter may be simulated directly at the desired points in time by methods described in that section. In addition, for the typical model parameters chosen in these experiments, we may conclude that the refined lognormal approximations are roughly comparable with 2δ - or 3δ -stepping Euler while the implementation of the latter is in many situations less efficient (see Section 6.3 for details).

REMARK 6.1 As the bias of different Libor approximations with respect to the terminal measure is caused by the approximation of the sum term in (6.3), we may expect that the larger this term is, the larger the bias will be. In particular, this sum term tends to zero when $t \downarrow t_0$ and when $i \uparrow n-1$. For instance, to estimate roughly which $L_i(T_i)$ has the largest bias in a specific lognormal approximation, we may approximate the sum term in (6.3) by replacing the L_j by their initial values (as in the (g)-approximation) and then obtain for the experiments in Section 6.2.2,

$$\ln \frac{L_i(T_i)}{L_i(0)} = \ln \frac{L_i(i\delta)}{L_i(0)} = \Sigma^{(i)} - \frac{1}{2}|\gamma|^2 i\delta + \int_0^{i\delta} \gamma_i^\top dW^{(n)}, \quad \text{where}$$

$$\Sigma^{(i)} \approx -|\gamma|^2 \frac{\delta^2 L(0)}{1 + \delta L(0)} \sum_{j=i+1}^{n-1} i \frac{e^{\beta(i-1)\alpha}}{e^{\beta(j-1)\alpha}} \quad (6.16)$$

and in the experiments, $L(0) = 0.061$, $\alpha = 0.9$, $\beta = 0.04$, $|\gamma| = 0.15$, $\delta = 0.25$, and $n = 61$. Since in practice α is typically close to 1, we now take $\alpha = 1$ in (6.16) for analytical tractability and thus obtain

$$|\Sigma^{(i)}| \approx |\gamma|^2 \frac{\delta^2 L(0)}{1 + \delta L(0)} \frac{1}{e^\beta - 1} i(1 - e^{-(n-1-i)\beta}).$$

Hence, for $\beta = 0$ (so, in fact, for a one-factor model), we have

$$|\Sigma^{(i_0^*)}| \approx |\gamma|^2 \frac{\delta^2 L(0)}{1 + \delta L(0)} i(n - i - 1),$$

which has a maximum

$$|\Sigma^{(i_0^*)}| \approx |\gamma|^2 \frac{\delta^2 L(0)}{1 + \delta L(0)} \frac{(n - 1)^2}{4} \quad \text{for } i = i_0^* \approx \frac{n - 1}{2},$$

whereas for $\beta \uparrow \infty$ (full de-correlation) it follows by elementary analysis that the $|\Sigma|$ -maximum, say $|\Sigma^{(i_\beta^*)}|$, goes to zero while $i_\beta^* \rightarrow n - 2$, non-decreasingly. In a particular situation, however, we can search numerically for i_β^* , and for the experiments in Section 6.2.2 we thus find $i_{0.04}^* \approx 37$, which is more or less consistent with the results listed in Tables 6.2–6.5. $\square$

6.3 Direct Simulation of Lognormal Approximations

The results given in Tables 6.2–6.5 show clearly that the lognormal models (0) and (g) are reasonable approximations and that the models ($g1'$), ($g1$), and ($g2$) are pretty good approximations to the solution of SDE (6.2). We now present direct techniques for lognormal approximations, in particular we illustrate an effective simulation of the (g)-approximation by a lognormal random field in Section 6.3.1.

The motivation for direct simulation methods is clear: In contrast to standard numerical solution of stochastic differential equations there is no need to take small time steps. Indeed, it is possible to construct the solution directly at the desired points in time, for example, at the points of the given tenor structure $0 < T_1 < T_2 < \dots < T_n$.² It will be shown in Section 6.4 that in many typical applications direct simulation methods take much less computing time.

6.3.1 Random Field Simulation of the (g)-approximation

Due to the simple correlation structure of the (g)-approximation it is possible to set up a lognormal random field model by a simulation technique as studied in Sabelfeld (1991) in a more general setting. We construct a lognormal random field whose first two statistical moments are consistent with those

²From now on, (T_i) is an arbitrary sequence of future time points, so T_i is not necessarily equal to $T_i = i\delta$ as in Section 6.2.2.

TABLE 6.1: Instantaneous correlations $\rho(\Delta L_i, \Delta L_j)$ between different Libors for $\alpha = 0.9$, $\beta = 0.04$

i, j	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60
4	1	0.88	0.79	0.70	0.63	0.57	0.51	0.46	0.42	0.38	0.34	0.31	0.28	0.26	0.23
8	0.88	1	0.89	0.80	0.71	0.64	0.58	0.52	0.47	0.43	0.39	0.35	0.32	0.29	0.26
12	0.79	0.89	1	0.89	0.80	0.72	0.65	0.59	0.53	0.48	0.43	0.39	0.36	0.32	0.29
16	0.70	0.80	0.89	1	0.90	0.81	0.73	0.66	0.59	0.54	0.49	0.44	0.40	0.36	0.33
20	0.63	0.71	0.80	0.90	1	0.90	0.81	0.73	0.66	0.60	0.54	0.49	0.44	0.40	0.37
24	0.57	0.64	0.72	0.81	0.90	1	0.90	0.81	0.73	0.66	0.60	0.55	0.49	0.45	0.41
28	0.51	0.58	0.65	0.73	0.81	0.90	1	0.90	0.82	0.74	0.67	0.61	0.55	0.50	0.45
32	0.46	0.52	0.59	0.66	0.73	0.81	0.90	1	0.90	0.82	0.74	0.67	0.61	0.55	0.50
36	0.42	0.47	0.53	0.59	0.66	0.73	0.82	0.90	1	0.90	0.82	0.74	0.67	0.61	0.56
40	0.38	0.43	0.48	0.54	0.60	0.66	0.74	0.82	0.90	1	0.91	0.82	0.74	0.68	0.61
44	0.34	0.39	0.43	0.49	0.54	0.60	0.67	0.74	0.82	0.91	1	0.91	0.82	0.75	0.68
48	0.31	0.35	0.39	0.44	0.49	0.55	0.61	0.67	0.74	0.82	0.91	1	0.91	0.82	0.75
52	0.28	0.32	0.36	0.40	0.44	0.49	0.55	0.61	0.67	0.74	0.82	0.91	1	0.91	0.82
56	0.26	0.29	0.32	0.36	0.40	0.45	0.50	0.55	0.61	0.68	0.75	0.82	0.91	1	0.91
60	0.23	0.26	0.29	0.33	0.37	0.41	0.45	0.50	0.56	0.61	0.68	0.75	0.82	0.91	1

TABLE 6.2: The cumulative distribution of the relative error ϵ_{12} , for 100 paths under different approximations

$100 \cdot \epsilon_{12}$	(0)	(g)	(g1')	(g1)	(g2)	δ st. E.	$2 - \delta$ st. E.
$\leq 0.01\%$	0	0	0	0	19	4	0
$\leq 0.02\%$	0	0	3	10	47	34	5
$\leq 0.03\%$	0	1	10	19	77	73	21
$\leq 0.04\%$	0	3	15	34	89	88	41
$\leq 0.05\%$	0	8	24	45	91	99	57
$\leq 0.06\%$	0	9	32	62	93	100	71
$\leq 0.07\%$	0	12	46	79	93	100	75
$\leq 0.08\%$	0	13	56	99	96	100	82
$\leq 0.09\%$	0	14	71	100	98	100	88
$\leq 0.1\%$	0	17	93	100	100	100	93
$\leq 0.2\%$	0	41	100	100	100	100	100
$\leq 0.3\%$	0	62	100	100	100	100	100
$\leq 0.4\%$	0	77	100	100	100	100	100
$\leq 0.5\%$	0	89	100	100	100	100	100
$\leq 0.6\%$	0	95	100	100	100	100	100
$\leq 0.7\%$	0	97	100	100	100	100	100
$\leq 0.8\%$	0	97	100	100	100	100	100
$\leq 0.9\%$	0	100	100	100	100	100	100
$\leq 1\%$	0	100	100	100	100	100	100
$\leq 2\%$	0	100	100	100	100	100	100
$\leq 3\%$	79	100	100	100	100	100	100
$\leq 4\%$	100	100	100	100	100	100	100

TABLE 6.3: The cumulative distribution of the relative error ϵ_{24} , for 100 paths under different approximations

$100 \cdot \epsilon_{24}$	(0)	(g)	(g1')	(g1)	(g2)	δ st. E.	3δ st. E.
$\leq 0.01\%$	0	0	0	0	3	0	0
$\leq 0.02\%$	0	0	0	0	4	17	0
$\leq 0.03\%$	0	0	2	0	14	50	0
$\leq 0.04\%$	0	0	2	4	20	71	6
$\leq 0.05\%$	0	0	4	9	30	91	10
$\leq 0.06\%$	0	0	10	11	37	96	15
$\leq 0.07\%$	0	2	16	16	47	99	23
$\leq 0.08\%$	0	2	17	22	51	99	32
$\leq 0.09\%$	0	2	19	24	57	99	48
$\leq 0.1\%$	0	3	22	29	66	100	53
$\leq 0.2\%$	0	16	59	93	92	100	92
$\leq 0.3\%$	0	22	100	98	96	100	99
$\leq 0.4\%$	0	34	100	100	99	100	100
$\leq 0.5\%$	0	43	100	100	100	100	100
$\leq 0.6\%$	0	47	100	100	100	100	100
$\leq 0.7\%$	0	60	100	100	100	100	100
$\leq 0.8\%$	0	69	100	100	100	100	100
$\leq 0.9\%$	0	73	100	100	100	100	100
$\leq 1\%$	0	75	100	100	100	100	100
$\leq 2\%$	0	98	100	100	100	100	100
$\leq 3\%$	0	100	100	100	100	100	100
$\leq 4\%$	21	100	100	100	100	100	100
$\leq 5\%$	65	100	100	100	100	100	100
$\leq 6\%$	91	100	100	100	100	100	100
$\leq 7\%$	98	100	100	100	100	100	100
$\leq 8\%$	100	100	100	100	100	100	100

TABLE 6.4: The cumulative distribution of the relative error ϵ_{36} , for 100 paths under different approximations

$100 \cdot \epsilon_{36}$	(0)	(g)	(g1')	(g1)	(g2)	δ st. E.	4δ st. E.
$\leq 0.01\%$	0	0	0	0	0	0	0
$\leq 0.02\%$	0	0	0	0	1	17	0
$\leq 0.03\%$	0	0	0	0	3	49	1
$\leq 0.04\%$	0	0	1	4	4	77	2
$\leq 0.05\%$	0	0	1	6	8	89	4
$\leq 0.06\%$	0	1	4	9	10	91	6
$\leq 0.07\%$	0	1	5	11	13	94	11
$\leq 0.08\%$	0	1	7	13	18	98	14
$\leq 0.09\%$	0	3	10	16	24	99	24
$\leq 0.1\%$	0	3	12	17	30	99	33
$\leq 0.2\%$	0	10	33	45	75	100	84
$\leq 0.3\%$	0	21	54	87	89	100	94
$\leq 0.4\%$	0	25	94	95	92	100	98
$\leq 0.5\%$	0	34	96	95	93	100	100
$\leq 0.6\%$	0	36	96	96	96	100	100
$\leq 0.7\%$	0	40	97	96	96	100	100
$\leq 0.8\%$	0	49	98	97	96	100	100
$\leq 0.9\%$	0	52	98	98	96	100	100
$\leq 1\%$	0	56	99	98	97	100	100
$\leq 2\%$	0	89	100	100	100	100	100
$\leq 3\%$	1	97	100	100	100	100	100
$\leq 4\%$	12	99	100	100	100	100	100
$\leq 5\%$	42	100	100	100	100	100	100
$\leq 6\%$	75	100	100	100	100	100	100
$\leq 7\%$	89	100	100	100	100	100	100
$\leq 8\%$	93	100	100	100	100	100	100
$\leq 9\%$	97	100	100	100	100	100	100
$\leq 10\%$	99	100	100	100	100	100	100

TABLE 6.5: The cumulative distribution of the relative error ϵ_{48} , for 100 paths under different approximations

$100 \cdot \epsilon_{48}$	(0)	(g)	(g1')	(g1)	(g2)	δ st. E.	4δ st. E.
$\leq 0.01\%$	0	0	0	0	0	6	0
$\leq 0.02\%$	0	0	0	0	0	55	0
$\leq 0.03\%$	0	0	1	0	0	82	0
$\leq 0.04\%$	0	0	2	1	6	90	4
$\leq 0.05\%$	0	0	2	4	9	95	12
$\leq 0.06\%$	0	0	3	4	13	97	21
$\leq 0.07\%$	0	0	7	8	14	97	31
$\leq 0.08\%$	0	0	10	12	20	98	41
$\leq 0.09\%$	0	0	15	19	21	98	52
$\leq 0.1\%$	0	0	16	22	26	99	67
$\leq 0.2\%$	0	8	40	46	71	100	89
$\leq 0.3\%$	0	17	74	85	83	100	97
$\leq 0.4\%$	0	22	93	93	91	100	99
$\leq 0.5\%$	0	30	95	94	92	100	100
$\leq 0.6\%$	0	35	95	95	95	100	100
$\leq 0.7\%$	0	40	96	95	95	100	100
$\leq 0.8\%$	0	41	96	96	95	100	100
$\leq 0.9\%$	0	45	96	96	96	100	100
$\leq 1\%$	0	54	96	96	96	100	100
$\leq 2\%$	2	90	100	100	100	100	100
$\leq 3\%$	19	97	100	100	100	100	100
$\leq 4\%$	55	98	100	100	100	100	100
$\leq 5\%$	79	100	100	100	100	100	100
$\leq 6\%$	88	100	100	100	100	100	100
$\leq 7\%$	95	100	100	100	100	100	100
$\leq 8\%$	97	100	100	100	100	100	100
$\leq 9\%$	99	100	100	100	100	100	100
$\leq 10\%$	100	100	100	100	100	100	100

of the (g) -approximation. We thus introduce a lognormal random model

$$\mathcal{L}^{(g)}(i, t) = \exp[\xi^{(g)}(i, t)] \quad (6.17)$$

in the measure P_n , where the $\xi^{(g)}(i, t)$, $i = 1, \dots, n-1$, $t_0 \leq t \leq T_i$, are Gaussian, such that mean and covariance structure coincide with that of $\ln(L_i^{(g)}(t)/L_i(t_0))$, $t_0 \leq t \leq T_i$, $i = 1, \dots, n-1$. Hence,

$$E_n \xi^{(g)}(i, t) = E_n \ln \frac{L_i^{(g)}(t)}{L_i(t_0)}, \quad (6.18)$$

$$E_n \xi^{(g)}(i_1, t_1) \xi^{(g)}(i_2, t_2) = E_n \left(\ln \frac{L_{i_1}^{(g)}(t_1)}{L_{i_1}(t_0)} \right) \left(\ln \frac{L_{i_2}^{(g)}(t_2)}{L_{i_2}(t_0)} \right). \quad (6.19)$$

From (6.11) we see that

$$\begin{aligned} E_n \xi^{(g)}(i, t) &=: \mu^{(g)}(i, t; t_0) \\ &= - \sum_{j=i+1}^{n-1} \frac{\delta_j L_j(t_0)}{1 + \delta_j L_j(t_0)} \int_{t_0}^t \gamma_i^\top \gamma_j(s) ds - \frac{1}{2} \int_{t_0}^t |\gamma_i|^2(s) ds, \end{aligned} \quad (6.20)$$

where in definition (6.20), t_0 is included as a fixed parameter in the expectation function $\mu^{(g)}$. Further, by (6.11) we have (with $\wedge$ denoting minimum),

$$E_n \xi^{(g)}(i_1, t_1) \xi^{(g)}(i_2, t_2) = \mu^{(g)}(i_1, t_1; t_0) \mu^{(g)}(i_2, t_2; t_0) + \text{cov}^{(g)}(i_1, i_2, t_0, t_1 \wedge t_2), \quad (6.21)$$

where

$$\text{cov}^{(g)}(i_1, i_2, s, t) := \int_s^t \gamma_{i_1}^\top \gamma_{i_2}(s) ds, \quad t_0 \leq s \leq t.$$

We now construct numerically the desired random field $\mathcal{L}^{(g)}(i, T_j)$, $1 \leq i < n$, $1 \leq j \leq i$. To do this, we could simulate the Gaussian vector with the given covariance structure by a conventional simulation technique; see for instance Sabelfeld (1991). However, the specific time correlation (6.21) resulting from the fact that $\xi^{(g)}$ has independent increments suggests a different simulation algorithm.

Indeed, in the first step, we simulate a $n-1$ -dimensional Gaussian vector $[\xi^{(g)}(1, T_1), \dots, \xi^{(g)}(n-1, T_1)]$ as

$$\xi^{(g)}(i, T_1) = \mu^{(g)}(i, T_1; t_0) + \sum_{k=1}^{K_1} h_{ik}^{(1)} \eta_k^{(1)}, \quad i = 1, \dots, n-1, \quad (6.22)$$

where the $(n-1) \times K_1$ matrix $[h_{ik}^{(1)}]$ satisfies a Cholesky decomposition

$$\sum_{k=1}^{K_1} h_{ik}^{(1)} h_{jk}^{(1)} = \text{cov}^{(g)}(i, j, t_0, T_1), \quad i, j = 1, \dots, n-1, \quad (6.23)$$

$h_{ik}^{(1)} = 0$ for $i > n - 1 - K_1 + k$, $\{\eta_k^{(1)}\}_{k=1}^{K_1}$ is a set of independent standard Gaussian random numbers, and K_1 is the rank of the covariance matrix (6.23). Note that, in general, $K_1 \leq n - 1$ and $K_1 = d$, the number of independent Brownian motions in the Libor model, in the case where the γ_i are time independent.

In the l -th step ($2 \leq l \leq n - 1$) we have

$$\begin{aligned} \xi^{(g)}(i, T_l) &= \xi^{(g)}(i, T_{l-1})\mu^{(g)}(i, T_l; t_0) - \mu^{(g)}(i, T_{l-1}; t_0) \\ &\quad + \sum_{k=1}^{K_l} h_{ik}^{(l)} \eta_k^{(l)}, \quad i = l, \dots, n - 1, \end{aligned} \quad (6.24)$$

where the $(n - l) \times K_l$ matrix $[h_{ik}^{(l)}]_{i,k=l,\dots,n-1}$ satisfies a decomposition

$$\sum_{k=1}^{K_l} h_{ik}^{(l)} h_{jk}^{(l)} = \text{cov}^{(g)}(i, j, T_{l-1}, T_l), \quad i, j = l, \dots, n - 1, \quad (6.25)$$

$h_{ik}^{(l)} = 0$ for $i > n - 1 - K_l + k$, $\{\eta_k^{(l)}\}_{k=1}^{K_l}$ is a set of independent standard Gaussian random numbers, and K_l is the rank of the covariance matrix (6.25). So, in general, $K_l \leq n - l$ and $K_l \leq \min(d, n - l)$ in the case where the γ_i are time independent.

After $n - 1$ steps we thus find

$$\begin{aligned} L_i^{(g)}(T_j) &= L_i(t_0)\mathcal{L}^{(g)}(i, T_j) \\ &= L_i(t_0)\exp[\xi^{(g)}(i, T_j)], \quad i = 1, \dots, n - 1, j = 1, \dots, i. \end{aligned} \quad (6.26)$$

Analogously, the same procedure could be easily carried out for the lognormal approximations (0). Indeed, the simulation formulae (6.22)–(6.26) remain the same, but for the (0)-approximation the functions $\mu^{(g)}$ and $\text{cov}^{(g)}$ should be replaced with

$$\mu^{(0)}(i, t; t_0) \equiv 0, \quad \text{and} \quad \text{cov}^{(0)}(i_1, i_2, s, t) = \text{cov}^{(g)}(i_1, i_2, s, t), \quad t_0 \leq s \leq t.$$

REMARK 6.2 Of course, in a random field Monte Carlo simulation all the Cholesky decompositions above can be computed outside of the simulation loop. In general, the cost of the Cholesky decomposition in the l -th step of the random field construction is $\mathcal{O}((n - l)^3)$, so the total cost of the different Cholesky decomposition is $\mathcal{O}(n^4)$. Moreover, in the case where the γ_i are *time independent* it is easily seen that, in fact, only one Cholesky decomposition has to be computed at a cost of $\mathcal{O}(n^3)$. However, for a detailed cost comparison of random field simulation to other direct simulation methods and Euler stepping SDE simulation, see Section 6.4.1. $\square$

6.3.2 Simulation of the $(g1')$, $(g1)$, and $(g2)$ -approximation

Let us consider the $(g1')$ -approximation by example, as the $(g1)$ and $(g2)$ can be treated analogously. For getting the $(g1')$ -approximation we have to plug (6.12), with the sum term ignored, into (6.13). Similar to (6.17) we then introduce a lognormal random model

$$\mathcal{L}^{(g1')}(i, t) = \exp[\xi^{(g1')}(i, t)] \quad (6.27)$$

in the measure P_n and define the expectation function

$$\begin{aligned} E_n \xi^{(g1')}(i, t) &= E_n \ln \frac{L_i^{(g1')}(t)}{L_i(t_0)} =: \mu^{(g1')}(i, t; t_0) \\ &= - \sum_{j=i+1}^{n-1} \delta_j L_j(t_0) \int_{t_0}^t \gamma_i^\top \gamma_j(s) ds - \frac{1}{2} \int_{t_0}^t |\gamma_i|^2(s) ds. \end{aligned}$$

We now derive from (6.13) by a Fubini argument,

$$\begin{aligned} \ln \frac{L_i^{(g1')}(t)}{L_i(t_0)} &= \mu^{(g1')}(i, t; t_0) \\ &+ \int_{t_0}^t \left(1 - \sum_{j=i+1}^{n-1} \delta_j L_j(t_0) \int_s^t \gamma_j^\top \gamma_i(u) du \right) \gamma_i^\top(s) dW^{(n)}(s), \end{aligned}$$

and thus obtain the covariance function,

$$\begin{aligned} \text{Cov}(\xi^{(g1')}(i_1, t_1), \xi^{(g1')}(i_2, t_2)) &= \int_{t_0}^{t_1 \wedge t_2} ds \gamma_{i_1}^\top \gamma_{i_2}(s) \times \\ &\left(1 - \sum_{j=i_1+1}^{n-1} \delta_j L_j(t_0) \int_s^{t_1} \gamma_j^\top \gamma_{i_1}(u) du \right) \left(1 - \sum_{k=i_2+1}^{n-1} \delta_k L_k(t_0) \int_s^{t_2} \gamma_k^\top \gamma_{i_2}(u) du \right). \end{aligned} \quad (6.28)$$

For the $(g1)$ and $(g2)$ -approximation one can derive similar approximations straightforwardly from (6.13), (6.14), respectively.

Unfortunately, the covariance functions of $(g1)$, $(g1')$, and $(g2)$ do not have the special structure that the (0) and (g) -approximation do, so a random field construction as in Section 6.3.1 does not work. For instance, the increment $\ln L_i(T_l) - \ln L_i(T_{l-1})$ is now in general correlated with $\ln L_i(T_{l-1})$ for $i \geq l-1$. However, it is possible to simulate the desired log-Libors simultaneously as one q -dimensional random vector ξ . Let the index set $\mathcal{I}$ be the collection of pairs (i, j) for which $L_i(T_j)$, $1 \leq j \leq i \leq n-1$ is to be simulated. So, q is equal to the number of elements of $\mathcal{I}$ and, for instance, $q = n(n-1)/2$ in case Libors over the whole tenor structure are required. Let further $\phi: \mathcal{I} \rightarrow \{1, \dots, q\}$ be an arbitrary bijection, then

$$\xi_{\phi(i,j)} := \ln \frac{L_i^{(\cdot)}(T_j)}{L_i(t_0)} = \mu^{(\cdot)}(i; t_0, T_j) + h_{\phi(i,j),1} \eta_1 + \dots + h_{\phi(i,j),K} \eta_K, \quad (i, j) \in \mathcal{I}, \quad (6.29)$$

where $(\cdot)$ stands for $(g1')$, $(g1)$, $(g2)$, respectively, the $q \times K$ matrix h satisfies a Cholesky decomposition

$$\sum_{k=1}^K h_{pk} h_{lk} = \text{cov}^{(\cdot)}(\xi_p, \xi_l) \quad p, l = 1, \dots, q, \quad (6.30)$$

$h_{pk} = 0$ for $p > q - K + k$, $\{\eta_k\}_{k=1}^K$ is a set of independent standard Gaussian random numbers, and K is the rank of the covariance matrix (6.30) which, for a specific lognormal approximation, is determined by its covariance structure, for instance, (6.28).

For a full tenor structure, hence for $\mathcal{O}(n^2)$ log-Libors, the Cholesky decomposition will now require a computational cost of $\mathcal{O}(n^6)$ and compared to this the cost of the drift terms can be ignored. It is important to note, however, that all these computations can be done outside of the simulation loop.

6.3.3 Cost Analysis of Euler SDE Simulation and Direct Simulation Methods

Here we give formulae for the cost of Euler SDE simulation, random field simulation of the (g) -approximation, and the direct simulation method for the other lognormal approximations. We disregard all computations which can be done outside of the Monte Carlo simulation loop, such as the computation of various Cholesky decompositions etc.

Let us suppose that we are faced with the simulation of

$$L_i(T_j), \quad 1 \leq j \leq m; \quad j \leq i \leq n-1, \quad (6.31)$$

for fixed m in the measure P_n .

Euler scheme for solving the log-Libor SDE system

$$dY_i = - \sum_{j=i+1}^{n-1} \frac{\delta_j e^{Y_j}}{1 + \delta_j e^{Y_j}} \gamma_i^\top \gamma_j dt - \frac{1}{2} |\gamma_i|^2 dt + \gamma_i^\top dW^{(n)}, \quad i = 1, \dots, n-1, \quad (6.32)$$

where $Y_i := \ln L_i(t)$. For the volatilities $\gamma_i = (\gamma_{i,1}, \dots, \gamma_{i,d})$, $1 \leq i \leq n-1$, we may assume that $\gamma_{i,k} = 0$ for $i > n-1-d+k$, and then it is not difficult to verify that the computation of a single Euler step from t to $t + \Delta t$, $t < T_1$,

incurs a cost of

$$\begin{aligned}
 \text{Cost}_{\text{Euler step}}(n, d, 1) = & \quad (6.33) \\
 & \overbrace{(n-2)(c_* + c_{\dot{*}} + c_+ + c_{\text{exp}}) + \frac{(n-2)(n-1)}{2}(c_* + c_+)}^{\text{Computation drift term}} \\
 & \overbrace{+ d(c_{\text{rand}} + c_*) + [(n-1-d)d + \frac{d^2+d}{2}](c_* + c_+)}^{\text{Computation noise term}} \\
 =: & (n-2)\tilde{c}_{\text{exp}} + \left[\frac{(n-2)(n-1)}{2} + nd - \frac{d^2+d}{2} \right] \tilde{c}_* + d\tilde{c}_{\text{rand}},
 \end{aligned}$$

where the cost of one addition, multiplication, division, exponentiation, and the generation of a standard Gaussian random number is denoted by c_+ , c_* , $c_{\dot{*}}$, c_{exp} , and c_{rand} , respectively. In practice $\tilde{c}_* \approx c_*$, $\tilde{c}_{\text{exp}} \approx c_{\text{exp}}$, $\tilde{c}_{\text{rand}} \approx c_{\text{rand}}$. From (6.33) it is obvious that we have in general for $T_{i-1} < t < T_i$, $i \geq 1$, (where $T_0 := t_0$)

$$\begin{aligned}
 \text{Cost}_{\text{Euler step}}(n, d, i) = & \quad (6.34) \\
 & (n-i-1)\tilde{c}_{\text{exp}} + \left[\frac{(n-i-1)(n-i)}{2} \right. \\
 & \left. + \max(n-i-d, 0)d + \frac{\min^2(d, n-i) + \min(d, n-i)}{2} \right] \tilde{c}_* \\
 & + \min(d, n-i)\tilde{c}_{\text{rand}} =: \mathcal{C}(d, n-i).
 \end{aligned}$$

REMARK 6.3 In our applications n is typically large (e.g., $n \approx 40, 60$) and therefore we could try to deal with asymptotic expression for the behaviour of $\mathcal{C}(d, k)$ for large k and certain d . However, we have to be careful; the cost of the exponential function and the Gaussian random number generator is considerably higher than the cost of a multiplication. For the compiler we used we found by experiment $c_{\text{exp}}/c_* \approx 25$, and with this compiler we found for the random number generator we used, $c_{\text{rand}}/c_* \approx 10$. So in a typical situation, for instance $k \approx 40, 60$, the term involving c_{exp} in (6.34) can not be ignored at all, and the same is true for the term involving c_{rand} when $d = n-1$ (a full factor model). Therefore, it is important to consider (6.34) for all k , rather than for $k \rightarrow \infty$ only. $\square$

Obviously, (6.34) yields

$$\begin{aligned}
 \mathcal{C}(d, k) &= (k-1)\tilde{c}_{\text{exp}} + k^2\tilde{c}_* + k\tilde{c}_{\text{rand}}, \quad k < d, \\
 \mathcal{C}(d, k) &= (k-1)\tilde{c}_{\text{exp}} + \left[\frac{(k-1)k}{2} + (k-d)d + \frac{d^2+d}{2} \right] \tilde{c}_* + d\tilde{c}_{\text{rand}}, \quad k \geq d.
 \end{aligned} \quad (6.35)$$

The numerical experiments in (6.2.2) have shown that, in practice, for a uniform tenor structure it is accurate enough to take time steps of order $\Delta t = \delta, 2\delta$ for time t up to T_1 and between two tenors T_i, T_{i+1} , we take δ_i for the Euler step size. So, it is clear that the total cost of a thus organized SDE simulation of one sample of the values (6.31) will be equal to

$$Cost_{SDE}(n, d, m) = \frac{T_1}{\Delta t} Cost_{Euler \text{ step}}(n, d, 1) + \sum_{i=2}^m Cost_{Euler \text{ step}}(n, d, i) + Cost_{Exp. \text{ calls}}^{(m)} \quad (6.36)$$

where $Cost_{Exp. \text{ calls}}^{(m)}$ takes into account the cost of the exponential calls at T_m to obtain Libors rather than log-Libors. For $i < m$ these exponential calls are already included in the first and second term of (6.36) as Libors at T_i are needed in the drift terms. Hence,

$$Cost_{SDE}(n, d, m) = \frac{T_1}{\Delta t} \mathcal{C}(d, n-1) + \sum_{k=n-m}^{n-2} \mathcal{C}(d, k) + (n-m)\tilde{c}_{exp} \quad (6.37)$$

with empty sums defined as zero. We derive from (6.35) and (6.37) by elementary algebra³ explicit expressions for the cost of the Euler method for $m > n-d$ and $m \leq n-d$, respectively,

$$\begin{aligned} Cost_{SDE}(n, d, m) = & \quad (6.38) \\ & \frac{T_1}{\Delta t} \left\{ (n-2)\tilde{c}_{exp} + \left[\frac{(n-2)(n-1)}{2} + nd - \frac{d^2+d}{2} \right] \tilde{c}_* + d\tilde{c}_{rand} \right\} \\ & + \left\{ (n-1)m - \frac{m^2+3m-4}{2} \right\} \tilde{c}_{exp} \\ & + \left\{ -\frac{n^3}{6} + \frac{2m+d-1}{2}n^2 - \frac{6m^2+6m+3d^2+6d-10}{6}n \right. \\ & \quad \left. + \frac{2m^3+3m^2+m+d^3+3d^2+2d-6}{6} \right\} \tilde{c}_* \\ & + \left\{ -\frac{n^2}{2} + \frac{2m+2d+1}{2}n - \frac{m^2+m+d^2+3d}{2} \right\} \tilde{c}_{rand}, \quad m > n-d \end{aligned}$$

³The expressions (6.38) and (6.39) can be checked easily with Mathematica or Maple, for instance.

and

$$\begin{aligned}
 \text{Cost}_{\text{SDE}}(n, d, m) = & \frac{T_1}{\Delta t} \left\{ (n-2)\tilde{c}_{\text{exp}} + \left[\frac{(n-2)(n-1)}{2} + nd - \frac{d^2+d}{2} \right] \tilde{c}_* + d\tilde{c}_{\text{rand}} \right\} \\
 & + \left\{ (n-1)m - \frac{m^2+3m-4}{2} \right\} \tilde{c}_{\text{exp}} + \left\{ \frac{m-1}{2}n^2 - \frac{m^2+2m-2md+2d-3}{2}n \right. \\
 & \quad \left. + \frac{m^3+3m^2+2m-6-3d^2m-3dm^2+3d^2+3d}{6} \right\} \tilde{c}_* \\
 & + (m-1)d\tilde{c}_{\text{rand}}, \quad m \leq n-d.
 \end{aligned} \tag{6.39}$$

The expressions (6.38) and (6.39) will be used later for a cost comparison of random field simulation and direct simulation of different lognormal approximations with Euler SDE simulation in various situations.

Random field simulation technique

Here we consider the general case where γ is time dependent and we thus have to take in (6.24), $K_l = n-l$ for $1 \leq l \leq n$, even if d is small. We choose the $(n-l) \times (n-l)$ -matrix $h^{(l)}$ in (6.24) as an upper-triangular matrix. Disregarding again pre-computation costs outside of the Monte Carlo loop, it follows from (6.24) that the cost of calculating one sample of the Libors (6.31) is given by

$$\begin{aligned}
 \text{Cost}_{\text{RFS}}(n, m) &= \sum_{l=1}^m \left\{ \frac{1}{2}(n-l)(n-l+1)\tilde{c}_* + (n-l)\tilde{c}_{\text{rand}} \right\} + \text{Cost}_{\text{Exp. calls}}^{(m,n)} \\
 &= \frac{1}{6}(3mn^2 - 3m^2n + m^3 - m)\tilde{c}_* + \frac{1}{2}m(2n-m-1)(\tilde{c}_{\text{rand}} + \tilde{c}_{\text{exp}}),
 \end{aligned}$$

where the term $\text{Cost}_{\text{Exp. calls}}^{(m,n)} = \frac{1}{2}m(2n-m-1)\tilde{c}_{\text{exp}}$ is taken into account because we need Libors rather than log-Libors.

REMARK 6.4 Let us consider a volatility structure which is piecewise constant on the interval $[T_1, T_n]$, hence $\gamma_i(s) = \gamma_i^{(p)}$ for $T_p \leq s < T_{p+1}$, $1 \leq p < n$. In this situation the first Cholesky decomposition in the random field construction (6.23) might have rank $n-1$, but for the Cholesky decompositions in step $2 \leq l < n$ we may take $K_l = \min(d, n-l)$. As a consequence, the random field simulation will be much faster in case d is small. Indeed, it is

easily verified that for the computation cost we now have

$$\begin{aligned}
 Cost_{\text{RFS}}^{\gamma \text{ p.c. } [T_1, T_n]}(n, d, m) &\leq \frac{1}{2}n(n-1)\tilde{c}_* + (n-1)\tilde{c}_{\text{rand}} + \\
 &\quad \sum_{l=2}^m [(n-l)d\tilde{c}_* + d\tilde{c}_{\text{rand}}] + Cost_{\text{Exp. calls}}^{(m,n)} \\
 &= \frac{1}{2}n(n-1)\tilde{c}_* + (n(m-1) - \frac{m^2 + m - 2}{2})d\tilde{c}_* \\
 &\quad + (n + md - d - 1)\tilde{c}_{\text{rand}} + \frac{1}{2}m(2n - m - 1)\tilde{c}_{\text{exp}}.
 \end{aligned} \tag{6.40}$$

Even for a full tenor structure $m = n - 1$, we have

$$\begin{aligned}
 Cost_{\text{RFS}}^{\gamma \text{ p.c. } [T_1, T_n]}(n, d, m = n - 1) &\leq (\frac{n^2 - n}{2} + \frac{n^2 - 3n + 2}{2}d)\tilde{c}_* \\
 &\quad + (n(d + 1) - 2d - 1)\tilde{c}_{\text{rand}} + \frac{1}{2}n(n - 1)\tilde{c}_{\text{exp}},
 \end{aligned} \tag{6.41}$$

and so the coefficients of $\tilde{c}_*$, $\tilde{c}_{\text{rand}}$, and $\tilde{c}_{\text{exp}}$ in (6.40) and (6.41) are for small d quadratic and linear in n , respectively, whereas the corresponding coefficients in the cost of the SDE simulation tend to be larger: For example, when $m = n - 1$ in (6.38) the coefficient of $\tilde{c}_*$ still contains $n^3/6$ and $\frac{T_1}{\Delta t} \frac{n^2}{2}$ for small d .

If, moreover, γ_i is constant on $[t_0, T_1[$, i.e., $\gamma_i(s) = \gamma_i^{(0)}$ for $t_0 \leq t < T_1$, a rank- d Cholesky decomposition applies to the $\xi^{(g)}(T_1)$ construction also and then we have

$$\begin{aligned}
 Cost_{\text{RFS}}^{\gamma \text{ p.c. } [t_0, T_n]}(n, d, m) &\leq \sum_{l=1}^m [(n-l)d\tilde{c}_* + d\tilde{c}_{\text{rand}}] + Cost_{\text{Exp. calls}}^{(m,n)} \\
 &= (nm - \frac{m^2 + m}{2})d\tilde{c}_* + md\tilde{c}_{\text{rand}} \\
 &\quad + \frac{1}{2}m(2n - m - 1)\tilde{c}_{\text{exp}},
 \end{aligned} \tag{6.42}$$

which means a next speed up with respect to Euler SDE simulation in the case where $d \ll n$; for example, compare (6.42) with (6.39) for $d = 1$. $\square$

Direct simulation of the (g_1') , (g_1) , and (g_2) -approximation

By taking $q = \sum_{j=1}^m (n - j) = \frac{1}{2}m(2n - m - 1)$ in (6.29) we see that with an upper-triangular h , for the full rank case $K = q$, the simulation of one sample

of (6.31) will cost inside of the loop:

$$\begin{aligned} \text{Cost}_{\text{DS}}(n, m) &= \frac{1}{2}q(q+1)\tilde{c}_* + q\tilde{c}_{\text{rand}} + \text{Cost}_{\text{Exp. calls}}^{(m,n)} \\ &= \left\{ \frac{m^2}{2}n^2 - \frac{m^3 + m^2 - m}{2}n + \frac{m(m+1)(m^2 + m - 2)}{8} \right\} \tilde{c}_* + \\ &\quad + \frac{1}{2}m(2n - m - 1)(\tilde{c}_{\text{rand}} + \tilde{c}_{\text{exp}}). \end{aligned} \quad (6.43)$$

It is clear that for $m = n - 1$ the first term will be of order $\mathcal{O}(n^4 c_*)$, and so direct simulation of the $(g1')$, $(g1)$, or $(g2)$ -approximation for a full tenor structure ($m = n - 1$) can only be recommended when n is not too large, whereas for larger n this simulation method is recommended for relatively small values of m . For example, $m = 1$ in (6.43) yields

$$\text{Cost}_{\text{DS}}(n, m = 1) = \frac{n^2 - n}{2}\tilde{c}_* + (n - 1)(\tilde{c}_{\text{rand}} + \tilde{c}_{\text{exp}}). \quad (6.44)$$

6.4 Efficiency Gain with Respect to SDE Simulation; an Optimal Simulation Program

6.4.1 Simulation Alternatives

Let us suppose that we want to do a Monte Carlo evaluation of a Libor derivative involving Libors specified in (6.31). Rather than full Euler-stepping from the starting date t_0 , it may be profitable to simulate $L(T_1)$ using one of the lognormal approximations and then, for instance, proceed with Euler stepping through the remaining tenors, provided, of course, that $L(T_1)$ is well approximated. Alternatively, as long as the (g) -approximation is tolerable, one may apply the random field simulation technique. It is to be expected that for longer dated products (i.e., larger T_1) and in particular when, additionally, m is small (e.g., $m = 1$ in the case of a European style derivative), both alternatives may yield a substantial efficiency gain. We will compute the order of this gain for a “full” tenor structure, i.e., $m = n - 1$, and the European case $m = 1$, where we distinguish between the multi-factor case $d = n - 1$ and the one factor case $d = 1$.

Alternative 1: Lognormal approximation of $L(T_1)$ followed by Euler stepping

As a first alternative to full Euler stepping we consider a simulation algorithm that first simulates with a proper lognormal approximation the Libors $L(T_1)$ and then proceeds with Euler δ -stepping. Using the results of Section 6.3.3 we may straightforwardly compute for this method the efficiency ratio

$$R_{\text{Eff}}^{(1)}(n, d, m) := \frac{\text{Cost}_{\text{SDE}[t_0, T_m]}}{\text{Cost}_{\text{DS}[t_0, T_1] \& \text{SDE}[T_2, T_m]}}$$

and obtain the following results.

Full structure, multi-factor

$$R_{\text{Eff}}^{(1)}(n, n-1, n-1) \approx \frac{(n \frac{T_1}{\Delta t} + \frac{n^2}{2})\tilde{c}_{\text{exp}} + (n \frac{T_1}{\Delta t} + \frac{n^2}{2})\tilde{c}_{\text{rand}} + (n^2 \frac{T_1}{\Delta t} + \frac{n^3}{3})\tilde{c}_*}{\frac{n^2}{2}\tilde{c}_{\text{exp}} + \frac{n^2}{2}\tilde{c}_{\text{rand}} + \frac{n^3}{3}\tilde{c}_*}$$

Full structure, one factor

$$R_{\text{Eff}}^{(1)}(n, 1, n-1) \approx \frac{(n \frac{T_1}{\Delta t} + \frac{n^2}{2})\tilde{c}_{\text{exp}} + (\frac{T_1}{\Delta t} + n)\tilde{c}_{\text{rand}} + (\frac{n^2}{2} \frac{T_1}{\Delta t} + \frac{n^3}{6})\tilde{c}_*}{\frac{n^2}{2}\tilde{c}_{\text{exp}} + 2n\tilde{c}_{\text{rand}} + \frac{n^3}{6}\tilde{c}_*}$$

European, multi-factor

$$R_{\text{Eff}}^{(1)}(n, n-1, 1) \approx \frac{(n \frac{T_1}{\Delta t} + n)\tilde{c}_{\text{exp}} + n \frac{T_1}{\Delta t} \tilde{c}_{\text{rand}} + n^2 \frac{T_1}{\Delta t} \tilde{c}_*}{n\tilde{c}_{\text{exp}} + n\tilde{c}_{\text{rand}} + \frac{n^2}{2}\tilde{c}_*}$$

European, one factor

$$R_{\text{Eff}}^{(1)}(n, 1, 1) \approx \frac{(n \frac{T_1}{\Delta t} + n)\tilde{c}_{\text{exp}} + \frac{T_1}{\Delta t} \tilde{c}_{\text{rand}} + \frac{n^2}{2} \frac{T_1}{\Delta t} \tilde{c}_*}{n\tilde{c}_{\text{exp}} + n\tilde{c}_{\text{rand}} + \frac{n^2}{2}\tilde{c}_*}$$

Hence, when $\frac{T_1}{\Delta t} \gtrsim n$ it is clear that this alternative will be substantially faster than full Euler stepping.

Alternative 2: Random field simulation

When volatilities are not too high or T_m is not too large the (g)-approximation might be acceptable and then, particularly for a full tenor structure ($m = n-1$), it might be profitable to use the random field simulation technique. The efficiency ratio

$$R_{\text{Eff}}^{(2)}(n, d, m) := \frac{\text{Cost}_{\text{SDE}_{[t_0, T_m]}}}{\text{Cost}_{\text{RFS}_{[T_1, T_m]}}}$$

may again be computed using Section 6.3.3 and for $m = n-1$ we have

Full structure, multi-factor

$$R_{\text{Eff}}^{(2)}(n, n-1, n-1) \approx \frac{(n \frac{T_1}{\Delta t} + \frac{n^2}{2})\tilde{c}_{\text{exp}} + (n \frac{T_1}{\Delta t} + \frac{n^2}{2})\tilde{c}_{\text{rand}} + (n^2 \frac{T_1}{\Delta t} + \frac{n^3}{3})\tilde{c}_*}{\frac{n^2}{2}\tilde{c}_{\text{exp}} + \frac{n^2}{2}\tilde{c}_{\text{rand}} + \frac{n^3}{6}\tilde{c}_*}$$

Full structure, one factor

$$R_{\text{Eff}}^{(2)}(n, 1, n-1) \approx \frac{(n \frac{T_1}{\Delta t} + \frac{n^2}{2})\tilde{c}_{\text{exp}} + (\frac{T_1}{\Delta t} + n)\tilde{c}_{\text{rand}} + (\frac{n^2}{2} \frac{T_1}{\Delta t} + \frac{n^3}{6})\tilde{c}_*}{\frac{n^2}{2}\tilde{c}_{\text{exp}} + \frac{n^2}{2}\tilde{c}_{\text{rand}} + \frac{n^3}{6}\tilde{c}_*},$$

from which we conclude that, in general, the efficiency gain with respect to the first simulation alternative is limited: If, in the full rank case, $n^3\tilde{c}_*$ is much larger than $n^2\tilde{c}_{\text{exp}}$, random field simulation is about two times faster as simulation alternative 1. However, for a low factor model with $\gamma(s)$ piece wise constant between T_1 and T_n , it follows from Remark 6.4 that random field

TABLE 6.6: Values of $R_{\text{Eff}}^{(1)}$ for Example 6.1
($\text{Cost}_{\text{SDE}} \times 10^4 \tilde{c}_*$)

Δt	$d = 1$	$d = 2$	$d = 40$
δ	3.66 (11.6)	3.64 (12.0)	3.83 (19.0)
2δ	2.30 (7.0)	2.29 (7.5)	2.39 (11.8)
3δ	1.84 (5.8)	1.84 (6.0)	1.90 (9.4)

simulation is really faster than lognormal approximation followed by Euler-stepping, and also faster than alternative 1. In the one factor case we then have

Full structure, one factor

$$R_{\text{Eff}}^{(2), \gamma \text{ p.c.}}(n, 1, n - 1) \approx \frac{(n \frac{T_1}{\Delta t} + \frac{n^2}{2}) \tilde{c}_{\text{exp}} + (\frac{T_1}{\Delta t} + n) \tilde{c}_{\text{rand}} + (\frac{n^2}{2} \frac{T_1}{\Delta t} + \frac{n^3}{6}) \tilde{c}_*}{\frac{n^2}{2} \tilde{c}_{\text{exp}} + 2n \tilde{c}_{\text{rand}} + n^2 \tilde{c}_*}.$$

For any problem in practice for which alternatives 1 or 2 applies it will be easy to compute or estimate from the expressions in Section 6.3.3 the efficiency gain with respect to Euler-stepping SDE simulation. However, instead of listing many possibilities, we present below two typical situations in which the proposed simulation alternatives improve on standard Euler-stepping.

Example 6.1

Suppose we have to price a relatively long term Libor option with $T_1 = 12$ years, $T_n = 22$ years, three-month periods, and where Libors over the full tenor structure are required for the valuation of the contract ($m = n - 1$). A product of this type is, for instance, the trigger swap; see Section 6.5. So, $\delta = 0.25$, $n = 41$, $m = 40$. For our C++ compiler the relative cost of a standard Gaussian random number and the exponential function is approximately $\tilde{c}_{\text{rand}}/\tilde{c}_*$, ≈ 10 and $\tilde{c}_{\text{exp}}/\tilde{c}_* \approx 25$, respectively.

The relative cost of full Euler δ -stepping from t_0 to T_n with respect to simulation alternative 1 ($(g2)$ -approximation of $L(T_1)$ and Euler δ -stepping from T_2 to T_n),

$$R_{\text{Eff}}^{(1)} := \frac{\text{Cost}_{\text{SDE}}_{[t_0, T_n]}}{\text{Cost}_{\text{DS}}_{[t_0, T_1] \& \text{SDE}}_{[T_2, T_n]}},$$

is computed from expressions in Section 6.3.3 and listed in Table 6.6 for a one- and two-factor model, a multi-factor model, and different Euler step sizes, $\Delta t = \delta, 2\delta, 3\delta$ on $[t_0, T_1]$. The absolute computation costs of the SDE method in terms of 10^4 multiplications are given in brackets.⁴ □

⁴The approximate expressions in this section give nearly the same results.

TABLE 6.7: Values of $R_{\text{EFF}}^{(1)}$ for Example 6.2
($\text{Cost}_{\text{SDE}} \times 10^4 \tilde{c}_*$)

Δt	$d = 1$	$d = 2$	$d = 40$
δ	39.5 (8.7)	40.6 (8.9)	64.8 (14.3)
2δ	19.9 (4.3)	20.6 (4.5)	32.6 (7.1)
3δ	13.4 (2.9)	13.8 (3.0)	21.9 (4.8)

Example 6.2

We consider the same problem as in Example 6.1 except that now only $L_i(T_1)$, $i = 1, \dots, n-1$, are required for pricing the product; so now $m = 1$, hence the European case. For instance, the swaption and the callable reverse floater are products of this type; see Section 6.5 for details. The computed cost ratios of direct simulation of the $(g1)$, $(g1')$, or $(g2)$ -approximation with respect to Euler-stepping SDE simulation are listed in Table 6.7 for different step sizes Δt and different d . It is clear that the efficiency gain in this case is tremendous. Note that in general when $m = 1$, the simulation costs inside the loop of the (g) -approximation and the $(g1)$, $(g1')$, or $(g2)$ -approximation coincide, and so a refined lognormal approximation as a $(g1')$, $(g1)$, or $(g2)$ -approximation should be preferred in any case. $\square$

6.4.2 An Optimal Simulation Program

Based on the experimental results with respect to the accuracy of different Libor approximations in Section (6.2.2) and the cost comparison between standard Euler δ -stepping and different alternative simulation methods in section (6.4.1), we now propose the following procedure for the price simulation of a derivative structure involving a system (6.31) of Libors, in a given (calibrated) Libor model.

Step I: Compare the accuracy of different lognormal approximation for the given Libor model by using a path simulation program as designed in Section 6.2.2.

Step (II): Decide which approximations are acceptable in view of the accuracy required for the derivative structure and then choose one of the following options:

Case 1: If the $(g1)$, $(g1')$, or $(g2)$ -approximation is tolerable up to $L(T_1)$ and the (g) -approximation is not suitable in the sense that Case 2 below does not apply, generate $L(T_1)$ by a direct simulation method (Section 6.3.2) and proceed to T_m with Euler δ -stepping.

Case 2: If the (g) -approximation is tolerable over all tenors, the number of factors, d , is small, and $\gamma(s)$ is piecewise constant be-

tween T_1 and T_m , then apply the random field simulation technique (Section 6.3.1).

Case 3: If Case 1 and 2 do not apply, SDE simulation by δ -stepping, or if necessary by smaller time steps, will be acceptable in almost all practical situations.

6.5 Practical Simulation Examples

We now present some test results on the valuation of the swaption and the trigger swap and discuss the pricing of a callable reverse floater.

6.5.1 European Swaption

Let us consider a standard European payer swaption with maturity T_1 , strike κ , and principal \$1, which gives the right to enter at T_1 into a swap contract with strike κ . Hence, by Proposition 1.2, the swaption value is given in the terminal measure P_n by

$$Swpn(t) := Swpn_{1,n;\kappa}(t) = B_n(t)E_n^{\mathcal{F}_t} \frac{(Swap_{1,n;\kappa}(T_1))^+}{B_n(T_1)}, \quad (6.45)$$

where $Swap_{1,n;\kappa}(T_1)$ denotes the value of the swap contract at T_1 . From (6.45) and the expression for the price of a standard swap contract (1.23), it follows by a little algebra that the swaption value can be represented as

$$Swpn(t) = B_n(t)E_n^{\mathcal{F}_t} \left[\sum_{j=1}^{n-1} \frac{B_{j+1}(T_1)}{B_n(T_1)} (L_j(T_1) - \kappa) \delta_j \right]^+. \quad (6.46)$$

In (6.46), the bond ratios $B_{j+1}(T_1)/B_n(T_1)$ can be expressed in Libors by

$$\frac{B_{j+1}(T_1)}{B_n(T_1)} = \prod_{i=j+1}^{n-1} (1 + \delta_i L_i(T_1)). \quad (6.47)$$

The expressions (6.46) and (6.47) can thus be used for simulation of swaption prices based on a large enough Monte Carlo sample of the Libor process. The initial bond value $B_n(t_0)$ has to be derived from the initial present value curve given at t_0 .

In the spirit of Example 6.2 we simulate the value of an at the money swaption using different Libor approximations and the results are listed in Table 6.8. To estimate the systematic error in the swaption value due to a particular approximation, the swaption value is simulated by the log-SDE

TABLE 6.8: Swaption values for different Libor approximations simulated under the same Wiener processes in base points (i.e., $\times 10^{-4}$)

"true value"	δ -step E.	3δ -step E.	($g2$)	($g1$)	($g1'$)	(g)	(0)
379.07	379.54	380.71	381.69	379.33	377.28	408.18	516.4
Sys. err. (%)	0.12	0.43	0.69	0.07	-0.47	7.7	36.2

$T_1 = 12$, $n = 41$, $\delta_i \equiv 0.25$, $L_i(0) \equiv 0.06045$, $|\gamma_i| \equiv 0.15$, $\alpha = 0.9$, $\beta = 0.04$, $\kappa = 0.06045$; Number of trajectories 50000, Monte Carlo error (1 standard deviation) ≈ 5 bp. $\approx 1.3\%$, SDEs simulated with $\Delta t = 0.05$

for this approximation, where the same Wiener processes are used for all approximations.

We note that although the ($g2$)-approximation according to Tables 6.2–6.5 seems to be the best path-wise lognormal approximation overall, it produces larger outliers than ($g1$) and ($g1'$). For this reason, apparently, the ($g2$)-approximation does not give an essentially better swaption approximation than the ($g1$) gives. Further we note that by using a much faster direct simulation method the "real time" Monte Carlo error can be considerably reduced by increasing the number of simulations. For instance, a run of 500000 direct simulations of the ($g1$)-approximation takes, for this example, about 5 seconds on a 1Ghz computer and gives a Monte Carlo error of about 0.4%, whereas from Table 6.8 (full rank column) we see that even a 3δ -stepping Euler method, which is comparable to ($g1'$) in accuracy, would have taken about 40 times longer.

6.5.2 Trigger Swap

Next we consider a knock-in version of a payer trigger swap with zero spread coupons (see Section 4.1.1 for the general trigger swap). For given trigger levels $K_1, \dots, K_n$ and strike κ , the contract is specified as follows. As soon as the discretely monitored spot Libor $L_i(T_i)$ crosses a trigger level K_i from below, the contract holder enters obligatory into a payer swap with fixed coupon κ for the remaining period $[T_i, T_n]$, settled at the tenor dates $T_{i+1}, \dots, T_n$. According to Proposition 1.2 and (1.23), the value of this trigger swap expressed in the P_n -measure is given by

$$Trswap(t) = B_n(t)E_n^{\mathcal{F}_t} \left[\frac{1}{B_n(T_\tau)} \left(1 - B_n(T_\tau) - \kappa \sum_{j=\tau}^{n-1} B_{j+1}(T_\tau) \delta_j \right) \right], \quad (6.48)$$

where τ , the trigger index, is given by $\tau := \min_{1 \leq p < n} \{p \mid L_p(T_p) > K_p\}$. For Monte Carlo simulation of the trigger swap we express the random variable

TABLE 6.9: Trigger swap values for different Libor approximations simulated under the same Wiener processes in base points (i.e., $\times 10^{-4}$)

"true value"	δ -step E.	(g2)	(g1)	(g1')	(g)	(0)
428.51	429.09	431.84	429.46	427.58	461.42	567.14
Sys. err. (%)	0.13	0.77	0.22	-0.22	7.6	32.3

$T_1 = 12$, $n = 41$, $\delta_i \equiv 0.25$, $L_i(0) \equiv 0.06045$, $|\gamma_i| \equiv 0.15$, $\alpha = 0.9$, $\beta = 0.04$, $\kappa = 0.06045$, $K_i \equiv 0.08$; Number of trajectories 25000, Monte Carlo error (1 standard deviation) ≈ 7 bp. $\approx 1.6\%$, SDEs simulated with $\Delta t = 0.05$

inside the expectation of (6.48) in Libors only to obtain the representation

$$Trswap(t) = B_n(t)E_n^{\mathcal{F}_t} \left[-1 + \prod_{i=\tau}^{n-1} (1 + \delta_i L_i(T_\tau)) - \kappa \sum_{j=\tau}^{n-1} \delta_j \prod_{i=j+1}^{n-1} (1 + \delta_i L_i(T_\tau)) \right]. \tag{6.49}$$

As an application of Example 6.1, we simulate the trigger swap value under different Libor approximations, just as in the swaption example (Section 6.5.1). The trigger levels are taken to be equal, $K_i \equiv 0.08$, and the results are given in Table 6.9. Again we see that the (g2)-approximation does not give an essentially better result than the (g1)- or (g1')-approximation for the same reason as in the swaption example. Regarding the rather large systematic error of the (g)-approximation over a 10-year time period for the example trigger swap above, we propose a direct simulation method as in Case 1, Section 6.4.2, by using the (g1)-approximation. On our 1GHz computer a run of 100000 simulations of the (full factor) Libor model using this method takes about two minutes and yields a tolerable Monte Carlo error of about 0.8%, whereas a δ -step Euler method takes about 4 times longer according to Table 6.6. Moreover, although the systematic error of the δ -step Euler method is smaller than the error of the (g1)-approximation, the latter error is still much smaller than the Monte Carlo error in this case and so the δ -step Euler method will not give an essentially better result.

6.5.3 Callable Reverse Floater

In particular, for products where Example 6.2 applies the efficiency gain of the proposed simulation method compared with Euler stepping is quite big. A realistic example of such a product is a T_1 -callable reverse floater, which is a European call option on a reverse floater at T_1 . Generally, reverse floaters occur in different variations; see Section 4.1.5 for further details and properties. Let us take, as a special example, $\kappa' = 0$ in the definition in Section 4.1.5. Then the cash-flows of the reverse floater are given by

$$C_{T_{i+1}} := \delta_i L_i(T_i) - \delta_i \max(\kappa - L_i(T_i), 0), \quad 1 \leq i \leq n - 1.$$

According to Section 4.1.5, the price of the reverse floater at time $t \leq T_1$ is given by

$$RF(t) = B_1(t) - B_n(t) - \sum_{i=1}^{n-1} B_{i+1}(t) E_{i+1}^{\mathcal{F}_t} [\delta_i (\kappa - L_i(T_i))^+], \quad (6.50)$$

and can be evaluated analytically in a Libor market model, where $L_i(T_i)$ is log-normally distributed under P_{i+1} . As a consequence, $RF(T_1)/B_n(t)$ may be expressed explicitly as a function of $L(T_1)$, and so it is possible to compute the price of the T_1 -callable reverse floater,

$$CRF(t) := B_n(t) E_n^{\mathcal{F}_t} \left[\frac{RF(T_1)^+}{B_n(T_1)} \right], \quad (6.51)$$

by Monte Carlo simulation of $L(T_1)$ in the terminal measure. Hence Example 6.2 applies. Moreover, the efficiency gain for this product will still be large for medium-term maturities T_1 . Of course, these conclusions apply also for the case $\kappa' \neq 0$.

6.6 Summarisation and Final Remarks

From Tables 6.2–6.5 in Section 6.2.2, and the Examples 6.5.1 and 6.5.2, we conclude that application of the lognormal (g)-approximation as proposed by Brace, Gatarek and Musiela (1997) in the valuation of long dated derivative structures may lead to intolerable errors in the option values. In Section 6.2 different lognormal approximation were constructed; in particular, besides the (0) and (g)-approximation, we considered the ($g1'$), ($g1$), and ($g2$)-approximations. Although the (g)-approximation is not reliable for longer-maturity structures, the refined lognormal approximations ($g1'$), ($g1$), and ($g2$) are in most cases still acceptable in the sense that the systematic errors in the option values caused by these approximations are well within a Monte Carlo error of about 1%, which in practice is thoroughly tolerable for over the counter options. In view of the larger outliers produced by the ($g2$)-approximation, however, the ($g1$)-approximation appears to be the best candidate in practice, and its implementation according to one of the simulation strategies outlined in Section 6.4.2 turns out to be very efficient compared to standard simulation of the log-Libor SDEs with time steps δ , 2δ , or 3δ . Moreover, the efficiency gain becomes very considerable when this simulation method is applied to products which fit into Example 6.2, in particular European-style derivatives such as the callable reverse floater. For European Libor products, the lognormal approximations can also be used for Monte Carlo evaluation of the concerning hedge parameters (see for instance Milstein & Schoenmakers (2002)). For practical applications we suggest the

implementation of a path comparison method as described in Section 6.2.2 as a measuring instrument that helps to decide in a particular situation whether a certain lognormal approximation is acceptable or not. Finally, it should be noted that in this chapter smile effects are not taken into account and the extension of the methods presented here to extended Libor market models (see for instance Section 1.3.4, Andersen & Andreasen (1998)), in order to incorporate volatility skews, would be an interesting subject of future research.